

RISK FACTOR DECOMPOSITION ANALYSIS

SUMMARY

This project takes a simulation of asset returns from a typical immature DB pension plan in 2011, over the period from June 2013 to December 2025 and decomposes the returns into various risk factors using ordinary least squares (OLS) and LASSO regression, with principal components analysis (PCA) also performed in Python for a more detailed analysis of explanatory variable variances.

The return series is generated using index proxies for various asset classes and the typical asset allocation taken from the PPF Purple Book 2011.

For the full DB pension plan funding simulator project that the series was drawn from, visit my GitHub page: <https://github.com/curingd/db-pension-plan-funding-simulator>

The key findings are that returns are mainly explained by exposures to credit and changes in inflation using the OLS model, or credit and a negative linear relationship with equity using the LASSO model. These results are not contradictory as OLS regressions of S&P 500 returns over high-yield bonds on credit factors and inflation changes on credit and S&P 500 factors show high significance, so whether the factors are framed as credit, equity, or inflation exposures over the 2013-25 period are a matter of preference.

DATA

Below are the primary set of factors used for regression:

Factor	Proxy	Data Source
Risk-free rate	Average SONIA	Bank of England
Swap spread	3-year gilt yield minus 3-year OIS	Bank of England
Yield curve slope	20-year minus 3-year gilt yield	Bank of England
Yield curve curvature (butterfly)	2 x 10-year minus 3-year minus 20-year gilt yield	Bank of England
Investment-grade credit spread	SLXX minus interpolated 5.56Y gilt yield (duration matched)	FMP, Bank of England
High-yield credit spread over investment-grade	GHYS.L minus SLXX	FMP
Equity over high-yield credit	S&P 500 returns minus GHYS.L	FMP, Robert Shiller
Equity uncertainty	VIX	FMP
Equity risk premium	S&P 500 E/P minus average SONIA	Robert Shiller, Bank of England
Inflation	Implied real 3-year gilt minus 3-year gilt yield	Bank of England
GBP currency hedge	Average SONIA minus EFRF	Bank of England, FRED

Figure 1: Table of primary risk factors for OLS

After inspection of plots of the asset returns time series and the above factors, the following second-order factors were then derived for regression:

Factor	Proxy	Data Source
Yield curve slope change	Change in 20-year minus 3-year gilt yield	Bank of England
Yield curve butterfly change	Change in 2 x 10-year minus 3-year minus 20-year gilt yield	Bank of England
Equity uncertainty change	Percentage change in VIX	FMP
Inflation change	Change in implied real 3-year gilt minus 3-year gilt yield	Bank of England

Figure 2: Table of secondary risk factors from differencing primary risk factors

Apart from the equity uncertainty measure (VIX), each of these factors are ex post and represent realised return factors, rather than ex ante market expectations (for example, investment grade and high-yield CDS spreads could have replaced the credit fund indices). This is consistent with the use of realised asset returns, rather than forward-looking projections.

ORDINARY LEAST SQUARES REGRESSION

OLS regression was performed on various factor combinations, checking for model and factor significance, and examining normal QQ plots for the distributions of the regression residuals. These are documented in the various tabs of *risk_factor_decomposition_ols.xlsx* and Figure 3 presents comparable summary statistics:

Model	Dependent variable	N	R ²	Adj. R ²	F	Sig. F	Intercept		
							Coef.	t-stat	p-val
Full	Asset Returns (%)	144	0.681	0.665	41.5	8.47E-31	0.5520	1.16	0.2472
SONIA, Credit, ERP	Asset Returns (%)	144	0.677	0.668	73.0	3.35E-33	0.2033	1.61	0.1100
SONIA, Credit, ERP, Inflation	Asset Returns (%)	144	0.687	0.676	60.6	3.90E-33	-0.7171	-1.54	0.1251
Credit, ERP, Inflation	Asset Returns (%)	144	0.681	0.672	74.1	1.68E-33	-0.5994	-1.30	0.1970
SONIA, Credit, Inflation	Asset Returns (%)	144	0.681	0.672	74.1	1.64E-33	-0.5483	-1.20	0.2316
Credit, Inflation	Asset Returns (%)	144	0.681	0.674	99.4	1.57E-34	-0.5754	-1.29	0.1986
Credit	Asset Returns (%)	144	0.670	0.666	143.4	1.05E-34	0.3505	3.88	0.0002
Credit, SP-HY	Asset Returns (%)	144	0.671	0.664	95.0	1.36E-33	0.3183	2.35	0.0204
SONIA, Swap, Credit	Asset Returns (%)	144	0.673	0.664	71.5	8.84E-33	0.2640	2.06	0.0417
Credit, VIX	Asset Returns (%)	144	0.688	0.682	103.0	2.95E-35	1.1133	3.93	0.0001
DSlope, DButterfly, Credit, VIX	Asset Returns (%)	144	0.689	0.678	61.1	2.68E-33	1.1198	3.89	0.0002
Credit, VIX, DInflation	Asset Returns (%)	144	0.695	0.687	79.3	6.62E-35	1.1011	3.91	0.0001
Credit, VIX, DVIX, DInflation	Asset Returns (%)	144	0.733	0.723	75.7	7.97E-38	0.8087	2.97	0.0036
Credit, DInflation, DVIX	Asset Returns (%)	144	0.729	0.721	93.5	2.01E-38	0.4466	5.27	0.0000
Auxiliary / diagnostic regressions (different dependent variable)									
S&P Returns on IG and HY	S&P 500 Returns - HY (%)	144	0.996	0.996	16464.2	7.95E-168	-0.1407	-13.21	0.0000
Dinflation on Credit, SP-HY	Inflation Change (%)	144	0.089	0.070	4.6	4.33E-03	0.0043	1.57	0.1197

Figure 3: OLS Regression Summary Statistics

The most statistically significant model, both in overall terms (lowest significance F) and with the most significant intercept (lowest p-value) is the regression of the asset returns on credit factors, changes in inflation and changes in implied equity uncertainty. The full statistical output is given in Figure 4 and this implies a model for asset returns given by the formula (rounded to 4 significant figures):

$$R_t = 0.4466 + 0.6604IGS_t + 0.2989HYS_t + 9.238DINF_t - 0.01838DVIX_t + \varepsilon_t$$

Where:

$$\begin{aligned} IGS_t &= \text{investment grade bond spread over gilts} \\ HYS_t &= \text{high-yield bond spread over investment grade bonds} \\ DINF_t &= \text{change in inflation} \\ DVIX_t &= \text{change in implied equity uncertainty} \end{aligned}$$

The regression could just as well have been specified in terms of equity (S&P 500 as a proxy) returns or the spread of equity returns over high-yield bonds: as the results of the bottom two regressions in Figure 1 show, there are significant linear relationships between equity returns, credit returns and changes in inflation over the 2013-25 period studied.

Note that the regression above is unusual in that there is no factor for a risk-free rate, apart from a negative contribution from the investment grade credit spread. When SONIA was included as a regression factor, it was repeatedly found to be insignificant and the problem is apparent from comparing the graph of SONIA returns to the asset returns (Figures 5 and 6): SONIA is clearly not covariance stationary, whereas asset returns do appear to be. The post-COVID rise in interest rates in 2022 is within the lookback period, ending a long period of historically low rates post GFC.

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.853816624							
R Square	0.729002828							
Adjusted R Square	0.721204348							
Standard Error	0.986654147							
Observations	144							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	4	364.0065049	91.00162622	93.4801202	2.0149E-38			
Residual	139	135.3146104	0.973486406					
Total	143	499.3211153						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	0.446565569	0.08469338	5.272732852	5.02421E-07	0.279111704	0.614019435	0.279111704	0.614019435
IG Credit Spread (%)	0.660449223	0.050740885	13.01611567	7.96972E-26	0.560125476	0.760772969	0.560125476	0.760772969
HY-IG Spread (%)	0.29893259	0.062649784	4.771486328	4.57066E-06	0.175062832	0.422802348	0.175062832	0.422802348
Inflation Change (%)	9.237945341	3.786927918	2.439429939	0.015970585	1.750515816	16.72537487	1.750515816	16.72537487
Change in VIX (%)	-0.018382199	0.003596035	-5.111795851	1.03621E-06	-0.0254922	-0.011272198	-0.0254922	-0.011272198

Figure 4: Summary statistics for OLS regression of asset returns on credit, inflation changes and VIX changes.

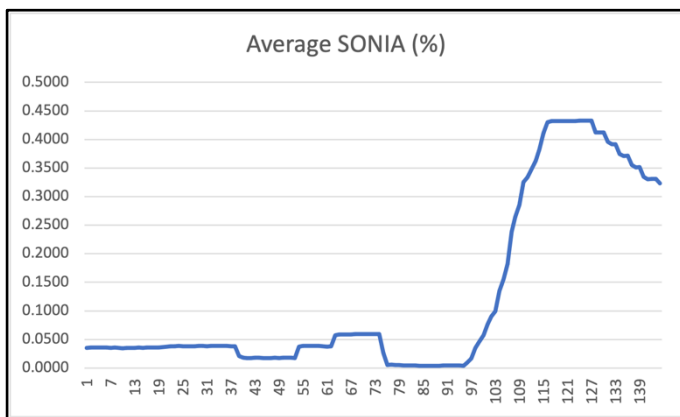


Figure 5: Average SONIA, June 2013 to December 2025

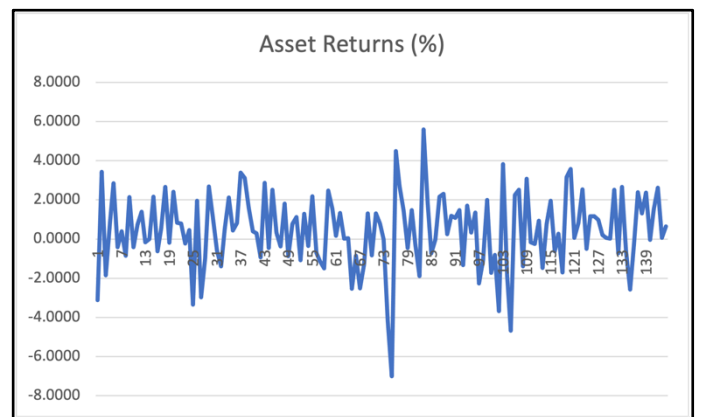


Figure 6: Asset returns, June 2013 to December 2025

The asset returns series's insensitivity to SONIA can be explained by both the predominant equity allocation of the returns (41.1% vs. 40.1% gilts), the higher volatility of equity relative to bonds and the global exposure inherent in the proxies used for private credit, equity, and hedge fund asset classes. From looking at the graph of the SONIA-EFFR spread (Figure 7), the lack of correlation between US and UK rates is also clear: positive correlation would appear as a near-flat graph.

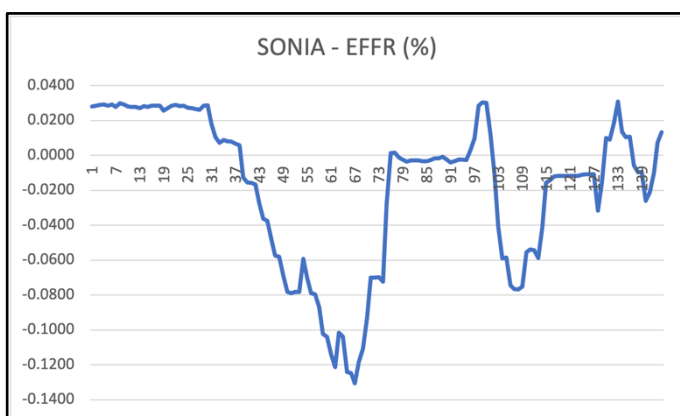


Figure 7: SONIA-EFFR base rate spread, June 2013 to December 2025

LASSO AND RIDGE REGRESSION

The process of identifying significant risk factors and reducing their number for parsimony was performed manually: an alternative is to use LASSO and/or ridge regression to penalise the sum of squares or the absolute sum of factor coefficients, respectively.

Of course, given the inconsistent measures used for the various risk factors, a prerequisite is to normalise each factor before regressing, so that large coefficients are not penalised purely because of the units of measurement of the corresponding factors.

Grid searching identified the optimal hyperparameters to use for ridge and LASSO regression and five-fold time splitting was used to separate training and test samples to identify the regression's predictive power within the 2013-25 period. Both aggregate in-sample and out-of-sample R^2 values and the changes in R^2 from fold to fold are in Figures 8 and 9. Both ridge and LASSO were compared to a near-unconstrained OLS regression on six principal components from principal components analysis (more on that later).

The headline: out-of-sample is where the truth comes out			
Model	In-sample R^2	CV R^2 (time-series)	CV-selected α
OLS	0.771	-0.974	—
Ridge	0.627	0.174	200.9
Lasso	0.674	0.543	0.305

Figure 8: in-sample and out-of-sample OLS/ridge/LASSO regression R^2 values

Model	Fold 1 (val Jun15– May17)	Fold 2 (Jun17– Dec19)	Fold 3 (Jan20– Dec21)	Fold 4 (Jan22– Dec23)	Fold 5 (Jan24– Dec25)	mean	worst
OLS	-1.69	+0.15	+0.66	-3.92	-0.07	-0.97	-3.92
Ridge	+0.29	-0.12	+0.57	-0.13	+0.26	+0.17	-0.13
Lasso	+0.60	+0.28	+0.70	+0.83	+0.32	+0.54	+0.28

Figure 9: per-fold out-of-sample changes in R^2 values for OLS/Ridge/Lasso regression

This points to the LASSO model as the most stable out-of-sample. Following on from the discussion of the asset returns' insensitivity to SONIA, the un-normalised Lasso regression avoided sensitivity to interest rate factors (see Figure 10 which documents the coefficients) and thus remained stable throughout the sample period.

	Ridge (raw)	Lasso (raw)	Ridge_intercept	Lasso_intercept
SONIA	0.03660501837	0	0.9612886444	0.3796699035
SONIA_EFFR_spread	2.004850958	0	0.9612886444	0.3796699035
slope	-0.3284841839	0	0.9612886444	0.3796699035
slope_change	2.132920484	0	0.9612886444	0.3796699035
butterfly	-0.9088681078	0	0.9612886444	0.3796699035
butterfly_change	-5.5244225	0	0.9612886444	0.3796699035
swap_spread	1.344337604	0	0.9612886444	0.3796699035
IG_gilt_spread	0.1595499565	0.2549746895	0.9612886444	0.3796699035
HY_IG_spread	-0.03647370941	0	0.9612886444	0.3796699035
inflation	0.2368541847	0	0.9612886444	0.3796699035
inflation_change	3.149717163	0	0.9612886444	0.3796699035
S&P_500_HY_spread	-0.2020328383	-0.2977533789	0.9612886444	0.3796699035
S&P_500_ERP	0.06897313113	0	0.9612886444	0.3796699035
VIX	-0.02828398525	0	0.9612886444	0.3796699035
VIX_change	-0.01006836715	-0.009736463363	0.9612886444	0.3796699035

Figure 10: un-normalised Ridge and LASSO regression coefficients

The next question to ask is whether the LASSO regression is a better fit than the most statistically significant model found through manually experimenting with factors in OLS. Figure 11 compares the R^2 , adjusted R^2 , AIC and BIC values of the credit/change-in-inflation/change-in-VIX model in Figure 3 with the LASSO model, and the LASSO model’s chosen factors re-run through OLS to give revised coefficients. The statistics clearly point to the original OLS model being the best explainer (in statistical terms) of asset returns:

Risk-factor regressions: in-sample model fit				
Model	R^2	Adjusted R^2	AIC	BIC
OLS (credit, dinflation, dvix)	0.7290	0.7212	411.70	429.51
Lasso (credit, equity, dvix)	0.6739	0.6669	436.34	451.19
Post-Lasso OLS (credit, equity, dvix)	0.7161	0.7100	416.39	431.24

n = 144. AIC/BIC: Gaussian ML, k = slopes + intercept + variance (identical convention across models). Lasso row = shrunk fit; Post-Lasso OLS = unbiased refit on the same three factors.

Figure 11: Comparison of summary statistics between manually chosen OLS, LASSO and post-LASSO OLS models.

CONCLUSIONS

To determine the primary drivers of asset returns, looking at raw coefficients from the 4-factor OLS model is insufficient: standardisation is required to factor out measurement assumptions and the distributions of the factors used for regression.

Writing each factor as $F_t = \mu_F + \sigma_F Z_t$ where μ_F and σ_F are the factor mean and standard deviation, respectively, gives the new regression coefficients as σ_F times the old ones for each factor, and the new intercept as $a + \sum_F \mu_F b_F$ where a is the old intercept and b_F is the old slope coefficient for factor F . This computation for the new coefficients is given in Figure 12.

Factor	Raw Coefficient	Factor Mean	Factor Standard Deviation	Standardised Coefficient
Intercept	0.446565569			0.444218728
IG Credit Spread (%)	0.660449223	0.047545544	2.386136491	1.575921991
HY-IG Spread (%)	0.29893259	0.118029769	1.921703697	0.574459864
Inflation Change (%)	9.237945341	0.000367133	0.02256576	0.208461253
Change in VIX (%)	-0.018382199	3.939830945	27.1965003	-0.499931484

Figure 12: standardised factor coefficients for 4-factor OLS regression

This reveals the largest driver of returns as the investment grade credit spread, followed by high-yield spreads over investment grade returns, then a negative relationship with equity uncertainty and a weaker relationship with inflation changes, despite the raw coefficient for inflation changes being of greatest magnitude. This conclusion is supported by corporate bonds being the largest weighted asset subclass from the PPF 2011 asset allocation (Figure 13, from the DB fund simulator):

	Asset Class	Overall Weight	Of Which	Weight Within	And	Weight Within	Subclass	Individual Weight	Adjusted (Sum to 1)		
2011	CASH	0.041					Cash	0.041	0.046		
	BONDS	0.401	Gilts	0.196	Short	0.121	Short Gilts	0.010	0.011		
							Medium Gilts	0.019	0.021		
			Corporate	0.443	Medium	0.239	Long Gilts	0.040	0.045		
							Corporate	0.178	0.199		
			Private	0			Private	0.000	0.000		
			IL	0.361	Long	0.512	Short IL	0.018	0.020		
							Medium IL	0.035	0.039		
			EQUITIES	0.411	UK	0.38			Long IL	0.074	0.083
	World	0.572			UK Equities	0.156			0.175		
	Private	0.048			World Equities	0.235			0.263		
					Private Equities	0.020			0.022		
	REAL ESTATE	0.044					Real Estate	0.044	0.049		
	HEDGE FUNDS	0.024					Hedge Funds	0.024	0.027		
								Sum:	0.892	1.000	

Figure 13: typical DB plan 2011 asset allocation, used to generate asset returns.

APPENDIX: PRINCIPAL COMPONENTS ANALYSIS (PCA)

Although it did not turn out as central to the analysis, PCA was performed on the standardised regression factors to offer insight into what factors may be selected for the regression and which may be discarded. The factor loadings for the first six principal components and component scree plots for total variation are given below alongside a correlation matrix for comparison.

The lack of “knee” to the scree plot, narrow range of factor loadings (-0.47 to 0.59) and predominance of weak pairwise correlations from the correlation matrix all point to a lack of conclusion regarding what factors to select based on factor covariances alone:

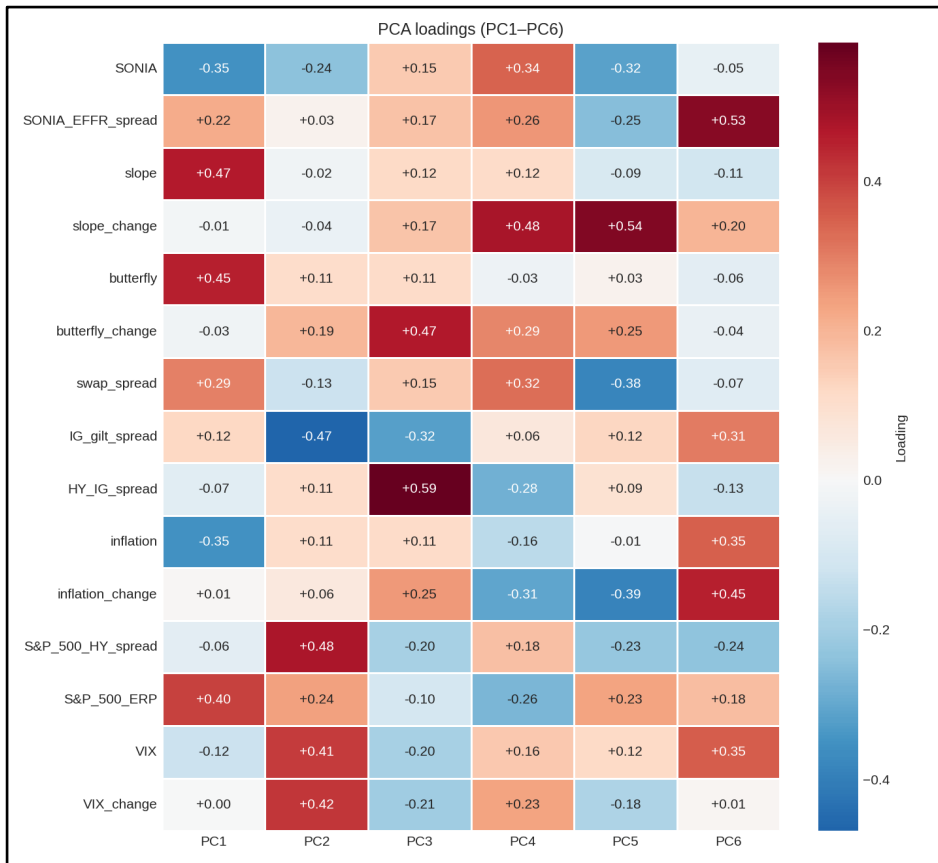


Figure 14: Factor loadings for PC1-6

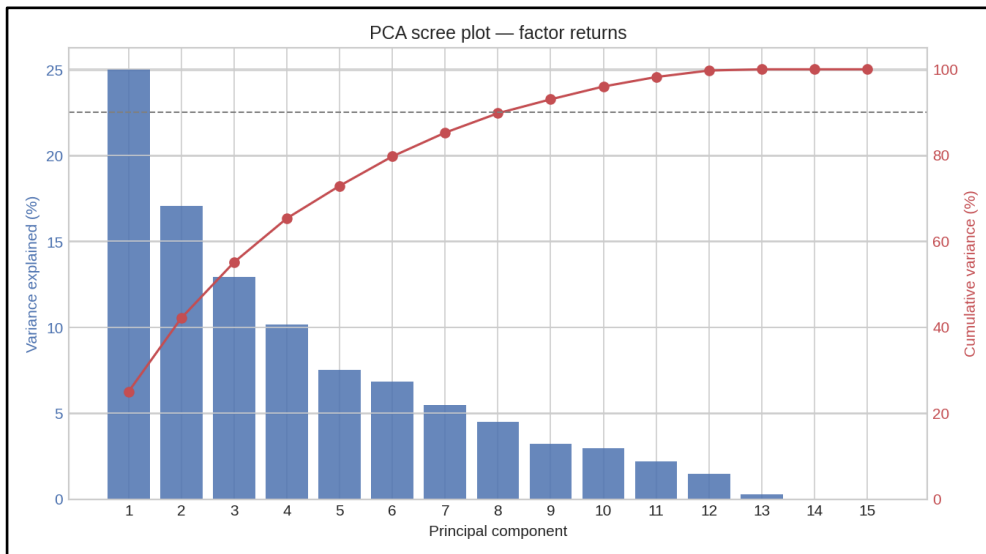


Figure 15: Scree plot for principal components with cumulative variation

	slope	butterfly	SONIA	SONIA_EFFR	swap_spread	slope_change	butterfly_change	IG_gilt	HY_IG	S&P_500_HY	inflation	inflation_change	VIX	VIX_change	S&P_500_ERP
slope	1.0000	0.8877	-0.4209	0.3965	0.5901	0.0761	0.0068	0.1186	-0.0431	-0.0840	-0.5745	0.0119	-0.3292	-0.0003	0.5263
butterfly	0.8877	1.0000	-0.5898	0.3822	0.3655	-0.0216	0.0891	-0.0077	0.0279	0.0056	-0.4039	0.0048	-0.2214	0.0578	0.6869
SONIA	-0.4209	-0.5898	1.0000	0.0030	-0.0417	0.1282	0.0744	-0.0005	0.0336	-0.0920	0.3483	-0.0666	-0.1223	-0.1324	-0.9619
SONIA_EFFR	0.3965	0.3822	0.0030	1.0000	0.3387	0.0701	0.1034	0.0451	-0.0073	-0.0543	-0.0931	0.0688	0.0623	-0.0002	0.2677
swap_spread	0.5901	0.3655	-0.0417	0.3387	1.0000	0.0364	0.1072	0.1674	-0.0763	-0.1198	-0.5138	0.0891	-0.2057	-0.0257	0.1245
slope_change	0.0761	-0.0216	0.1282	0.0701	0.0364	1.0000	0.3521	0.0838	-0.0172	-0.0936	0.0018	-0.1240	0.0718	-0.0079	-0.1079
butterfly_change	0.0068	0.0891	0.0744	0.1034	0.1072	0.3521	1.0000	-0.3875	0.3912	0.0805	0.0751	0.0562	0.1033	0.0808	-0.0500
IG_gilt	0.1186	-0.0077	-0.0005	0.0451	0.1674	0.0838	-0.3875	1.0000	-0.6071	-0.6180	-0.2718	-0.1223	-0.2840	-0.2945	0.0111
HY_IG	-0.0431	0.0279	0.0336	-0.0073	-0.0763	-0.0172	0.3912	-0.6071	1.0000	-0.2474	0.1948	0.2433	-0.0825	-0.1728	-0.0367
S&P_500_HY	-0.0840	0.0056	-0.0920	-0.0543	-0.1198	-0.0936	0.0805	-0.6180	-0.2474	1.0000	0.1181	-0.0799	0.4339	0.5370	0.0774
inflation	-0.5745	-0.4039	0.3483	-0.0931	-0.5138	0.0018	0.0751	-0.2718	0.1948	0.1181	1.0000	0.1864	0.2058	-0.0084	-0.3659
inflation_change	0.0119	0.0048	-0.0666	0.0688	0.0891	-0.1240	0.0562	-0.1223	0.2433	-0.0799	0.1864	1.0000	-0.0559	0.0157	0.0781
VIX	-0.3292	-0.2214	-0.1223	0.0623	-0.2057	0.0718	0.1033	-0.2840	-0.0825	0.4339	0.2058	-0.0559	1.0000	0.4711	0.1238
VIX_change	-0.0003	0.0578	-0.1324	-0.0002	-0.0257	-0.0079	0.0808	-0.2945	-0.1728	0.5370	-0.0084	0.0157	0.4711	1.0000	0.1280
S&P_500_ERP	0.5263	0.6869	-0.9619	0.2677	0.1245	-0.1079	-0.0500	0.0111	-0.0367	0.0774	-0.3659	0.0781	0.1238	0.1280	1.0000

Figure 16: Correlation matrix of all factors

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